

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12411703
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Ryman; Paul Adam et al.

Virtualized applications on managed virtual machines

Abstract

Examples of enterprise management using managed virtual machines for virtualized applications are described. In some examples, a managed virtual machine is executed on a host device and enrolled with a management service. A virtualized application drive is stored in the managed virtual machine. A launch agent executed within the managed virtual machine detects an application access request corresponding to user interaction with a host operating system of the host device. The application volumes launch agent accesses the virtualized application drive and launches the virtualized application corresponding to the application access request.

Inventors: Ryman; Paul Adam (Beecroft, AU), Parry-Barwick; Stephen Jonathan (Sidney, AU), Choudhary; Jairam (Palo Alto, CA), Passi; Arun (San Ramon, CA)

Applicant: Omnissa, LLC (Mountain View, CA)

Family ID: 85774799

Assignee: Omnissa, LLC (Mountain View, CA)

Appl. No.: 17/494072

Filed: October 05, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230106879 A1	Apr. 06, 2023

Publication Classification

Int. Cl.: G06F9/455 (20180101); G06F9/54 (20060101)

U.S. Cl.:

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8495323	12/2012	Tatavarty	711/6	G06F 3/0622
10652241	12/2019	Bendersky	N/A	H04W 12/30
2014/0181844	12/2013	Von Bokern	719/321	G06F 9/4411
2017/0371519	12/2016	Beveridge	N/A	G06F 9/45558
2020/0026540	12/2019	Venkatesh	N/A	G06F 9/5077
2020/0150987	12/2019	Nair	N/A	G06F 12/1081
2021/0042050	12/2020	Schauer	N/A	G06F 3/0647
2022/0066808	12/2021	Tsirkin	N/A	H04L 63/0263

Primary Examiner: Aquino; Wynuel S

Attorney, Agent or Firm: Kim & Stewart LLP

Background/Summary

BACKGROUND

(1) Personal devices are being used for access to enterprise computing systems and resources with greater frequency. More and more companies are now looking to incorporate “bring your own device” (BYOD) concepts for access to enterprise computing systems. With the growing number of BYOD devices, various challenges can arise in management scenarios. For example, the enterprise may use a management service to protect access to enterprise data and enterprise-related functionalities. However, a device may or may not be managed or subject to restrictions, policies, and other management configurations promulgated by the enterprise management. Access to the capabilities of a device can be managed through the administration of the suite of applications based on a set of management policies, compliance rules, and other configuration data which is defined and enforced by the management service. Since a user of a BYOD device can take the device on vacation or other locations without network access, the device may be used for an extended period of time without contacting the management service, causing issues with existing technologies.

(2) Application delivery can also be problematic for BYOD scenarios. Files and registry keys remain on a device, file type associations are not reverted properly, services and resources shared with other applications are incorrectly removed, services are not properly unregistered, application data and log files are not removed, or an uninstallation process fails without proper completion. Often re-installing the same application has issues with these leftover artifacts. This can be unacceptable for enterprise and end users alike.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale, with emphasis instead being placed upon clearly illustrating the principles of the disclosure. In the drawings, like reference numerals designate corresponding parts throughout the several views.

(2) FIG. 1 illustrates an example networked environment for distributing and managing managed virtual machines and virtualized applications, according to various examples described herein.

(3) FIG. 2 illustrates an example of packaging a managed virtual machine package and provisioning the managed virtual machine with virtualized applications, according to various examples described herein.

(4) FIG. 3 illustrates a flowchart performed by components of the networked environment, according to various examples described herein.

(5) FIG. 4 illustrates another flowchart performed by components of the networked environment, according to various examples described herein.

DETAILED DESCRIPTION

(6) The present disclosure relates to enterprise management using managed virtual machines (VMs) and virtualized applications. As outlined above, personal devices are being used to access enterprise applications, data, and other resources with greater frequency. The use of these personal devices can be associated with productivity gains and cost savings. The concept “bring your own device” (BYOD) for access to enterprise resources faces a number of issues, including the provision for offline access. Whether a device is provided by the enterprise or the end user, devices are more often operated from home and other locations that are remote from an enterprise local area network. Application delivery can be problematic for BYOD scenarios. Files and registry keys remain on a device, file type associations are not reverted properly, services and resources shared with other applications are incorrectly removed, services are not properly unregistered, application data and log files are not removed, or an uninstallation process fails without proper completion. Re-installing the same application has issues with these leftover artifacts. This can be unacceptable for enterprise and end users alike. However, the present disclosure provides a more effective solution that can provide both greater flexibility and isolation for virtualized application delivery in enterprise environments.

(7) FIG. 1 illustrates an example networked environment **100** for enterprise management using managed VMs **122** according to various examples described herein. The networked environment **100** includes a computing environment **103** and several host devices **106** in communication through a network **111**.

(8) The computing environment **103** can be embodied as one or more computers, computing devices, or computing systems. In certain embodiments, the computing environment **103** can include one or more computing devices arranged, for example, in one or more servers or computer banks. The computing device or devices can be located at a single installation site or distributed among different geographical locations. The computing environment **103** can include a plurality of computing devices that together embody a hosted computing resource, a grid computing resource, or another distributed computing arrangement. In some cases, the computing environment **103** can be embodied as an elastic computing resource where an allotted capacity of processing, network, storage, or other computing-related resources vary over time. As further described below, the computing environment **103** can also be embodied, in part, as certain functional or logical (e.g., computer-readable instructions) elements or modules as described herein.

(9) The computing environment **103** can operate as an environment for mobile device management or a unified endpoint management (UEM) platform that can manage the host devices **106**. In that

context, the computing environment **103** includes a data store **110**. The computing environment **103** can also execute a management service **120** and an identity provider **121**. The data store **110** includes areas in memory for the storage of management data **112**, policies **113**, profiles **114**, compliance status **115**, user data **117**, enterprise resources **119**, managed VMs **122**, and virtualized applications **124**, among other types of data. Enterprise resources **119** can include the virtualized applications **124**, as well as other enterprise data, enterprise-provided applications, and other resources.

(10) The management service **120** can operate as a UEM platform that can manage both the managed VMs **122**, as well as the host devices **106**, that are enrolled with the management service **120** for management. However, the management service **120** can manage the managed VMs **122** even if the host device **106** is unmanaged. An unmanaged host device **106** can include host devices **106** that are not enrolled with the management service **120**, and lack management-service-enforced policies, restrictions, profiles, or other management-service-enforced configurations for the host device **106** or the host operating system **151**. The management service **120** can be accessible over a public wide area network (WAN) such as the Internet.

(11) Management data **112** can represent information about managed VMs **122** and host devices **106** that are enrolled as managed devices with the management service **120**, as well as information about unmanaged host devices **106** on which enrolled managed VMs **122** are executed as guests.

(12) The management data **112** can include a unique VM identifier for the managed VM **122**, a device identifier of the host device **106** on which it is executed, certificates associated with the managed VM **122** or the host device **106**, a user identifier identifying the user account with which the host device **106** or managed VM **122** is linked, authentication tokens provided by the identity provider **121**, configuration profiles **114** and policies **113** assigned to the host device **106** and/or the managed VM **122**, and other information regarding management. The management data **112** can also include host specific information such as a virtual or physical trusted platform module (vTPM).

(13) The management data **112** can also include a compliance status **115** associated with the managed host device **106** and/or the managed VM **122**. The compliance status **115** can identify which rules are violated by the host device **106**, as well as any that are violated by the managed VM **122**. For example, the host device **106** may have been taken outside of a specified geofence defined for the host device **106** or the managed VM **122**. In another example, the managed VM **122** may fail a signature verification or a hash check of a hash of all or a portion of the managed VM **122**.

(14) User data **117** represents information about users who have user accounts in the enterprise. These users can also have one or more host devices **106** and one or more managed VMs **122** that are enrolled as managed entities with the management service **120**. User data **117** can include authentication data, and information about third-party services **107** with which the user is assigned an account.

(15) While host devices **106** can remain unenrolled, in some cases, the host devices **106** can enroll with the management service **120** for mobile device management services. To begin enrollment, the management service **120** can identify and authenticate one of the host devices **106** and store data related to the host device **106** in the management data **112** for later reference. In some cases, the management service **120** (or a management component, an application, or a component executing on the host device **106**) can also be registered as a device administrator (at least in part) of the host device **106**, permitting the management service **120** to configure and manage certain operating aspects of the host device **106**.

(16) Once the host device **106** is enrolled for device management by the management service **120**, the management service **120** can direct the installation of various software components or applications on the host device **106**. The software components can be configured on the host device **106** at the direction of the management service **120**. Such software components can include, for

example, applications, resources, libraries, and other related components.

(17) The management service **120** can transmit compliance rules including profiles **114**, policies **113**, and other configuration data for execution by and/or enforcement by a host management component on the host device **106**. In that context, during or after enrollment, the management service **120** can retrieve or generate a set of management policies, compliance rules, and configuration data for the host device **106** and transfer those policies, rules, and data to the host device **106** for reference by the host operating system **151** and certain applications executing on the host device **106**. The host management component or agent can be executed on the host device **106**, and can retrieve and install compliance rules and configuration data for execution and/or enforcement on the host device **106**. In other cases, the host device **106** can be unmanaged by the management service, and can lack policies, compliance rules, and configuration data that are enforced for the host user context.

(18) The management service **120** can also transmit compliance rules, policies **113**, profiles **114**, and configuration data for execution by and/or enforcement by a guest management component of the managed VM **122**. In that context, during or after enrollment, the management service **120** can retrieve or generate a set of management policies, compliance rules, and configuration data for the managed VM **122** and transfer those policies, rules, and data to the guest management component of the managed VM **122** for reference by a guest operating system and certain applications of the managed VM **122**. A guest management component or agent executed as part of the managed VM **122** can retrieve and install compliance rules and configuration data for execution and/or enforcement on the managed VM **122**.

(19) The management service **120** can also provide a management console as an engine and console interface for device management of the host devices **106**. An information technology administrator or user, for example, can view, administer, and update the management policies, compliance rules, and configuration data on the host devices **106** using the management console. The policies, rules, and configuration data can be collectively administered for several of the host devices **106** by organizing the host devices **106** into several different groups or categories of devices according to enterprise or other factors or considerations. The management service **120** can ensure compliance of the host device **106** in conjunction with a management component and other instructions executed by the computing environment **103** and the host device **106**.

(20) The identity provider **121** can provide single sign-on or identity management capabilities for an enterprise, whether a user is interacting with a user interface of the managed VM **122**, or a user interface of the host device **106**. The identity provider **121** can allow users to authenticate his or her identity to obtain an authentication token that can be provided to a third-party service **107**, which can include a data analysis service, and other services. The identity provider **121** can utilize OAuth, security assertion mark-up language (SAML), or other single sign-on methodologies. The identity provider **121** and management service **120** can communicate so that the management service **120** can revoke or authorize access to various services for users in the enterprise based on status of a host device **106** assigned to the user. The identity provider **121** can also rely on user data **117** in the data store **110**. In some examples, the identity provider **121** can rely upon a separate source of user data in a separate data store.

(21) A third-party service **107** can be embodied as one or more computers, computing devices, or computing systems. Like the computing environment **103**, a third-party service **107** can include one or more computing devices arranged, for example, in one or more servers or computer banks. The computing device or devices can be located at a single installation site or distributed among different geographical locations. Each of the third-party services **107** can include a plurality of computing devices that together embody a hosted computing resource, a grid computing resource, or other distributed computing arrangements. The third-party services **107** can also be embodied, in part, as certain functional or logical (e.g., computer-readable instructions) elements or modules as described herein. A third-party service **107** can be provided by an enterprise to its users. The third-

party service **107** can federate its authentication for users of the enterprise to the identity provider **121**. Alternatively, the third-party services **107** can be provided by the computing environment **103**, and can be part of the management service **120**.

(22) The management service **120** can use the third-party service **107** to deliver and distribute the virtualized applications **124**. In some examples, the virtualized applications **124** can be stored using the third-party service **107** rather than the management service **120**. In other cases, the virtualized applications **124** can receive the virtualized applications **124** from the management service **120**.

The third-party service **107** can include a content delivery network, peer to peer network, and other delivery and distribution services. In some cases, the content delivery network, peer to peer network, and other delivery and distribution services can be part of the management service **120**.

(23) The host devices **106** can be representative of one or more host devices **106**. Each host device **106** can be embodied as any computing device, processing circuit, or processor based device or system, including those in the form of a desktop computer, a laptop computer, a tablet computer, a personal digital assistant, a cellular telephone, a wearable computing device, or a set-top box, among other example computing devices and systems. Depending upon its primary purpose or function, for example, the host devices **106** can include various peripheral devices or components. The peripheral devices can include input or communications devices or modules, such as keyboards, keypads, touch pads, touch screens, microphones, cameras, wireless communications modules (e.g., infra-red, WI-FI, or BLUETOOTH®), buttons, switches, or sensors. The peripheral devices can also include a display, indicator lights, speakers, global positioning system (GPS) circuitry, accelerometers, gyroscopes, or other peripheral devices depending upon the primary purpose or function of the host devices **106**.

(24) An example host device **106** can also be enrolled by the management service **120** for device management. A host management component can be installed on a host device **106** to locally manage the device on behalf of the remotely executed management service **120**. The management component can be installed with elevated privileges or be effectuated through the host operating system **151** application programming interfaces (APIs) to manage the host device **106** on behalf of the management service **120**. The management component can have the authority to manage data on the host device **106**, install, remove, or disable certain applications, or install policies **113**, profiles **114**, such as VPN certificates, Wi-Fi profiles, email profiles, etc.

(25) The network **111** can include the Internet, intranets, extranets, wide area networks (WANs), local area networks (LANs), wired networks, wireless networks, cable networks, satellite networks, other suitable networks, or any combinations thereof. As one example, the computing environment **103** and the host devices **106** can be respectively coupled to one or more public or private LANs or WANs and, in turn, to the Internet for communication of data among each other. Although not shown in FIG. **1**, the network **111** can also include communicative connections to any number and type of network hosts or devices, such as website servers, file servers, cloud computing resources, databases, data stores, and any other network or computing architectures.

(26) In the networked environment **100**, the computing environment **103**, the host devices **106**, and other services can communicate data among each other over the network **111** using one or more network transfer protocols or interconnect frameworks, such as hypertext transfer protocol (HTTP), simple object access protocol (SOAP), representational state transfer (REST), real-time transport protocol (RTP), real time streaming protocol (RTSP), real time messaging protocol (RTMP), user datagram protocol (UDP), internet protocol (IP), transmission control protocol (TCP), other protocols and interconnect frameworks, and combinations thereof.

(27) The managed VM **122** can include a managed VM configuration file, VM configuration files, a non-volatile memory state file, virtual disk files, log files, as well as additional files and data. The managed VM **122** can include a guest operating system and a number of applications, certificates, Wi-Fi profiles, email profiles, and other data. The managed VM **122** can be enrolled with, and managed using, the management service **120**. The managed VM **122** can be preconfigured to

communicate with one or more management endpoints of the management service **120**. The managed VM **122** can include the certificates, credentials, and a management endpoint address to enable this communication. In addition, the managed VM **122** can provide an endpoint to which the management service **120** can transmit or push commands that can be performed by a management agent, service, or component executed within the managed VM **122**. A management agent of the managed VM **122** can also periodically check in with the management service **120** to provide operational information, as well as to retrieve commands from a command queue for the managed VM **122**.

(28) The managed VM **122** can utilize guest identity coupling with the host device **106** identity. For example, a hostname-detecting instruction for the managed VM **122** can be specified as part of the authoring and packaging process. In some cases, an identity of the host device **106** can be identified and can be used as an identity of the managed VM **122**. Alternatively, the identity of the host device **106** can be modified or used in combination with other identifiers, and the modified version of the host identity can be used as an identity of the managed VM **122**.

(29) The managed VM configuration file can include a textual or other configuration file, which can store settings chosen during creation of the VM through a console user interface of the management service **120**, or another VM settings editor. This can include a .cfg, .vmx, or another file extension. The managed VM configuration file can include file properties that are not included in a typical VM or its configuration file. For example, the managed VM configuration file can include managed VM configuration parameters, among other parameters. The NVRAM file can be a nonvolatile RAM file that can include items that cannot be stored in volatile memory, such as vTPM state and encryption keys, as well as the BIOS state data.

(30) These files can change infrequently and are relatively small in size relative to the whole managed VM **122**. The VM password or credentials can be automatically generated and encrypted and stored in a user login keychain for MacOS® or alternatively protected using a data protection API for Windows®. This provides a high key strength and simplicity to the user. In traditional models, if the user forgot their password, the content of the VM could be lost. This either leads to weak passwords or lost data.

(31) The management service **120** can include a packaging tool that packages a base VM into a managed VM package for a managed VM **122**. While the packaging tool can be accessed through a console user interface of the management service **120**, the packaging tool can also be installed and executed on another computing environment such as an administrator's personal computing device, a host device **106**, or another device. The packaging tool can modify a VM base image into a managed VM **122** by writing a managed VM configuration file into the VM base image. This can include modifying a VM configuration file of the VM base image by writing managed virtual machine configuration parameters into the VM configuration file.

(32) The management service **120** can include a provisioning tool that provisions a managed VM **122** on a host device **106**. While the provisioning tool can be stored by the management service **120**, the provisioning tool can also be installed and executed on a host device **106**. The provisioning tool can also be included in a management component or agent installed on the host device **106** for implementing commands provided by the management service **120**.

(33) The provisioning tool can initiate and implement a provisioning process. The provisioning process can use a managed VM package to install, configure, and launch a managed VM **122**. The provisioning tool can generate and implement command line arguments as well as scripts, API calls and operating system calls. The provisioning tool can: Extract the VM files from the managed VM package to a predetermined folder location. Verify a digital signature included with the managed VM package, including comparing hashes and verifying that a signer certificate is valid, trusted and not revoked. In some cases, if the package was not signed, no verification is performed. Add the managed VM **122** to the host desktop hypervisor **203** inventory of VMs. Notify the user that the VM is available to use. Provide a user interface element that lists all managed VMs **122** on the host

device **106**.

(34) Once the provisioning process is completed, a user or provisioning tool can cause the host desktop hypervisor **203** (FIG. 2) to execute the managed VM **122** for the first time. If the auto-add vTPM option is set to software in the managed VM configuration file, then the host desktop hypervisor **203** can add or create a vTPM device to the managed VM **122** using partial encryption and automatic key generation as described. If a user interface locking feature is enabled based on the editability status for editing settings of the managed virtual machine through a host desktop hypervisor **203**, the user will not be able to change a predetermined set (or any) of VM settings from the user interface of the host desktop hypervisor **203**. The managed VM **122** can be accessed in an out of box experience (OOBE) process if the sysprep or other audit mode instructions have been completed. The enrollment grace period timer can start once the managed VM **122** is executed for the first time and the OOBE process has started. A user can complete the OOBE and begin to use the managed VM **122**.

(35) A virtualized application **124** can include a virtualized version of any application utilized for enterprise purposes. Virtualized applications **124** can include Win32 applications and other types of applications. The virtualized application **124** can include a virtual application volume. In some examples, a virtualized application **124** can refer to an application that is provided by mounting a virtual disk containing the application to “layer” applications over a shared operating system kernel. In some examples, the operating system kernel used for the virtualized application **124** is the guest operating system kernel of the managed virtual machine **122**, rather than the host operating system kernel of the host device **106**. As a result, the managed virtual machine **122** can provide a secure container for all virtualized applications **124**. Further, since the virtualized application **124** is launched from a mounted virtual disk file, and write operations are siloed in a predetermined non-operating system write volume, it also does not affect the underlying guest operating system once closed.

(36) The virtualized applications **124**, which can include Win32 apps, can also be created and containerized using a lightweight OS-based kernel-level virtualization based on Hyper-V, such as Windows® Sandbox. With an addition or extension to share the host device **106** system files and memory pages with the host device **106** using a copy-on-write approach and integration with a host OS **151** scheduler. This can provide the benefit of full kernel virtualization but with a light memory and storage footprint. By using a method like instant clones, the boot time of a Windows® Sandbox container implementation is also near instantaneous. This implementation can lack of persistence between reboots, all writes during the session are lost, making it only suitable for testing out an untrusted app or accessing a potentially harmful web app in a protected environment. However, persistence can be achieved by directing write operations to a persistent nonvolatile writeable volume.

(37) The management service **120** can include a console user interface through which an administrator or other user can design the managed VM package and its managed VM **122**. In order to prepare the managed VM **122**, the management service **120** can generate a user interface through which a user can author a generalized Windows® or other VM desktop image. The VM base image can include the guest operating system as well as any applications, services, operating system features and language packs that are required for the base image. The user can select applications, certificates, Wi-Fi profiles, email profiles, data stores, network drives, and data that can be included in the managed VM **122**.

(38) The management service **120** can also validate that the VM base image for the managed VM **122** meets the managed VM rules that include packaging rules and Data Loss Prevention (DLP) rules for managed desktop VMs **122**. The managed VM rules can include requirements as well as recommendations.

(39) Managed VM **122** rules can specify requirements for the base image of the managed desktop VMs **122**. In one nonlimiting example where a VMware® Hypervisor is used, the requirements can

include: Not a linked clone No snapshots or linked clone VM derived from the VM No virtual disk files (e.g., .VMDK files) that are not dedicated to the VM (virtual disk files must be dedicated to the VM) The VM must not be marked as a template The VM working directory must be a current directory containing the VMX Minimum VM hardware version (e.g., >=14) Windows® 10 or another acceptable guest operating system Firmware type must be UEFI Only boot from HDD is permitted (no network or CD-ROM booting) Network devices must be bridged or NAT only (host-only and custom adapters not permitted) CD/DVD devices must be pass-through or “raw” devices only (i.e., not mapped to an ISO file) Floppy drives devices are not permitted Parallel port and virtual printer devices are not permitted Serial port devices are not permitted Pass-through SCSI devices are not permitted All HDD devices must map to virtual disk files (“raw” device pass-through is not permitted) Virtual disk files must exist in the same directory, to limit the chance of packaging a shared virtual disk file Camera devices are not permitted USB devices other than Hub and HID devices are not permitted Pass-through to host sound card devices are not permitted Shared folders must be disabled Option to snapshot the VM on power off must be disabled The AutoProtect rolling snapshotting feature must be disabled VNC connections must be disabled All appliance view settings including port must not be set Proxy app support either host-to-guest or guest-to-host must be disabled

(40) Recommendations for the managed VM rules can include: Enable secure boot Do not share Bluetooth devices with the host Disable guest isolation drag-and-drop Disable guest isolation copy-and-paste Do not synchronize guest time with the host Do not use Tool scripts Do not gather VM debug information Do not clean up disks after VM shutdown Enable Full or Partial Disk Encryption for the Managed VM **122**

(41) The management service **120** can also sign and package the managed VM **122** into a managed VM **122** in a format that can be deployed with any of: product cost and life cycle management (PCLM), system center configuration manager (SCCM), active directory (AD) or other products; a UEM or mobile device management (MDM), such as the management service **120** or a legacy management service; manual or enterprise script-based deployment using a removable drive, a file share, an FTP, or similar systems.

(42) A packaging tool can package a base VM into a managed VM **122**. The packaging tool can generate and implement command line arguments as well as scripts, API calls, and operating system calls. The packaging tool can: Provide a user interface through which a powered-down VM can be selected. For example, a .vmx file of a VM that has been designed through the user interface, or another VM can be selected for distribution as a managed VM **122**. Perform checks on the base VM to ensure an acceptable configuration for a managed VM **122** based on the managed VM rules outlined above. Provide a user interface through which a managed VM enterprise identifier can be entered, selected, or otherwise added to the VM .vmx file. The managed VM enterprise identifier can be associated with a set of enterprise-specific policies, baselines, profiles, applications, certificates, and other enterprise resources **119** to be included in a user context configuration of the managed VM **122**. Provide a user interface through which an enterprise display name can be entered, selected, or otherwise added to the VM .vmx file. Provide a user interface through which a user group identifier can be entered, selected, or otherwise added to the VM .vmx file. The user group can be associated with a set of user group-specific policies, baselines, profiles, applications, certificates, and other enterprise resources **119** to be included in a user context configuration of the managed VM **122**. Generate a digital signature based on an admin-provided certificate such as an enterprise certificate or a management service certificate. The single digital signature or multiple digital signatures can be generated based on the certificate and one or more of: the managed VM configuration file (or a subset of protected configuration settings within the managed VM configuration file), the virtual disk .vmdk file or files, and the NVRAM file. Provide a user interface through which a user can enter or select an enrollment grace period or period of time within which the managed VM **122** must enroll, re-enroll, check in, or otherwise

contact the management service **120** to maintain enrollment Provide a user interface through which a user can specify a cryptographic method that indicates whether a hardware-based crypto-processor, or a software-based or virtualized crypto-processor is to be used. The packaging tool can include a user interface through which a user can create, install and uninstall command-line arguments and scripts to use for PCLM, MDM, or manual deployment and provisioning.

(43) The managed VM configuration file can include protected parameters that are identified by the packaging tool at packaging time. These protected parameters can include a selected set of the managed VM enterprise identifier, the enterprise display name, the user group identifier, the grace period, the security level or cryptographic method, and others as discussed.

(44) The packaging tool can generate the managed VM **122** to include a .ppkg file and an unattend.xml file. The .ppkg file and an unattend.xml file can be copied to the managed VM **122**. The .ppkg file can include the files of the managed VM **122** as well as enterprise- and user-group-specific policies, baselines, profiles, applications, certificates, and other enterprise resources **119**. The files of the managed VM **122** can include a managed VM configuration file, an NVRAM file, virtual disk (.vmdk) files, a log file, as well as additional files and data. The managed VM **122**, or the .ppkg, can also include a signature and a number of additional files that are not part of the managed VM **122**, or the signature. For example, a signing verification or signed parameter file, as well as other configuration files. The packaging tool can identify or receive the enterprise resources **119** from the management service **120** based on a request that includes the managed VM enterprise identifier and the user group identifier.

(45) Once packaged, the managed VM **122** can be distributed to the host device **106** by the management service **120**, a third-party or legacy PCLM, SCCM, AD, UEM, MDM, or other management service, manually, or using a script. The managed VM **122** can be accessed during deployment through a removable drive connected to the host device **106**, a file share accessed by the host device **106**, an FTP, or similar systems. The management service **120** can, in some cases, distribute the managed VM **122** to the host device **106** once a host managed VM service and the host desktop hypervisor are installed.

(46) A provisioning tool can be included with the host managed VM service **209** (FIG. 2) or can be provided separately. In some cases, the provisioning tool can automatically select the .ppkg file and an unattend.xml file from the managed VM **122**, or can provide a user interface through which a user can select the .ppkg file and an unattend.xml file, and a user interface element to install the managed VM **122**.

(47) The provisioning tool can verify authenticity and integrity of the managed VM **122** based on a managed VM digital signature that is based on a management service **120** or enterprise certificate. Once the authenticity and integrity of managed VM **122** is verified, the host managed VM service **209** can setup host-device-specific VM settings including adding a vTPM device with encryption and generating a unique VM identifier for the managed desktop VM **122**.

(48) The provisioning tool can also add the managed VM **122** to a VM inventory of the host desktop hypervisor of the host device **106**. The host managed VM service can then notify a user of the host device **106** that the managed VM **122** is ready for use.

(49) FIG. 2 can illustrate an example of packaging a managed virtual machine package and provisioning a managed VM **122** with virtualized applications **124** a host device **106** using components of the networked environment **100**. The managed VM **122** can be used as a provisioning package to install the managed VM **122** on the host device **106**.

(50) The host device **106** can include a host desktop hypervisor **203**, a host managed VM service **209**, and a managed VM **122**. The host managed VM service **209** can be used in concert with a guest managed VM service **210** for management of the managed VM **122**. The host managed VM service **209** can be used primarily for communications through the IPC channel **211** between the managed VM **122** environment and the host environment. The host managed VM service **209** can lack a network communication ability, and does not communicate with the management service

120. As a result, the host device **106** can be considered a management agentless device that lacks a management agent **212** that is configured to authenticate and communicate with the management service **120**.

(51) The managed VM **122** can include the guest managed VM service **210**, a guest management agent **212**, a virtual application agent **215** or another launch redirection agent, and virtualized applications **124a . . . 124n**. Virtualized applications **124a . . . 124n** can be launched from mounted application virtual disk volumes **224a . . . 224n**. VM installed applications **218** can be installed to a guest operating system volume **236**. The managed VM **122** can also include a guest operating system **221** and a guest operating system volume **236** where the guest operating system **221** is installed.

(52) The managed VM **122** can also include certificates, Wi-Fi profiles, email profiles, and other guest user context configurations according to the profiles **114** and policies **113** associated with a user group or a user of the host device **106**. The managed VM **122** can also include enterprise data that is accessed by the applications and guest operating system **221** of the managed VM **122**. The managed VM **122** can further include links or addresses to create and access network drives and network services that provide the enterprise data.

(53) The managed virtual machine **122** within the host device **106** is managed using the management agent **212** or UEM agent. As a result, enterprise software and applications can be managed to comply with policies **113**, profiles **114**, and other configurations, while the underlying host device **106** is not actively managed for compliance with the policies **113** and profiles **114**. For example, many users of host devices **106** do not opt to use full volume encryption, which can be required by policies **113** and profiles **114** enforced by the management agent **212**. However, the entirety of the managed virtual machine **122** can be enforced to use full volume encryption, including the application virtual disk volumes **224a . . . 224n**, the writeable volume or volumes **234**, the guest operating system volume **236**, as well as all data corresponding to the managed virtual machine **122**. While this data is stored using the underlying host hardware **206** that is allocated to the managed virtual machine **122**, the host operating system **151**, the personal applications **251a . . . 251n**, and other data for the host environment can be stored in an unencrypted state.

(54) Full or partial disk encryption can be provided using an operating system-based encryption such as BitLocker®, or non-operating-system-based encryption such as VeraCrypt, TrueCrypt, and so on. As a result, while personal applications **251a . . . 251n** can be unencrypted, the application virtual disk volumes **224a . . . 224n** can be encrypted using operating system-based encryption. Full disk encryption can be enforced for the managed VM **122**. Even if the host device **106** is encrypted, the managed VM **122** can benefit from a cloud-based encryption key escrow that maintains the recovery keys for enrolled managed devices and managed VMs **122**. The management service **120** can securely store and provide secured access to recover keys escrowed in the cloud-based encryption key escrow and accessible over the Internet, unlike local enterprise intranet based solutions. The disk encryption configurations can be applied equally to any virtual application package **251**. For example, application specific virtual disk volumes created for virtualized applications **124**, and writeable volume **234** virtual disks can be encrypted as they are created.

(55) Using virtualized applications **124a . . . 124n** can provide a number of security features over personal applications **251a . . . 251n**, as well as VM installed applications **218**. For example, virtualized applications **124a . . . 124n** mount virtual disks or application virtual disk volumes **224a . . . 224n** containing the corresponding application to “layer” applications over the shared guest operating system **221** kernel. Disk and registry write operations by the application that would otherwise be made to the local guest OS volume **236** can be redirected to a special writeable volume or volumes **234**, thus preventing changes to the operating system or other installed applications.

(56) This can be used to avoid app compatibility conflicts and unwanted dependencies common

between different installed applications. Uninstalling a traditional Win32 application, for example, can be troublesome—often files and registry keys remain, file type associations are not reverted properly, services and resources shared with other applications are incorrectly removed, services are not properly unregistered, application data and log files not removed, or the uninstallation process fails silently without proper completion. Re-installing the same application can result in issues with leftover artifacts. The use of technologies like virtualized applications **124a . . . 124n** for legacy Win32 applications solves these issues. Virtualized applications **124a . . . 124n** can allow OS images for the guest OS **221** to be independently updated from the virtualized applications **124a . . . 124n**.

(57) The managed VM **122** can provide the isolation, management, security and data loss prevention (DLP) requirements of an enterprise and can provide an environment for hosting work-related applications and data. The management service **120** can provide services to manage the managed VM **122** as it would a typical managed device, but without impacting the host OS **151** and the user's personal environment. The management service **120** can deliver security policies **113** and profiles **114** to limit how data can be shared between the host device **106** environment and the managed VM **122**, such as disabling cut-and-paste between host and guest or disabling sharing host files with the guest. The application layering using virtualized applications **124** provides the additional level of isolation between virtualized applications **124a . . . 124n** and the managed VM **122** guest OS **221** itself to ensure application compatibility, and to maintain the guest OS **221** clean and unaffected by the virtualized applications **124**.

(58) The use of virtualized applications **124** within the managed VM **122** provides both a level of process-level containerization within the managed VM **122** itself and kernel-level virtualization for a stricter form of isolation of work applications such as the virtualized applications **124** and data from the user's personal applications **251a . . . 251n**, as well as data on the host operating system **151**. This means the host operating system **151** is unutilized and unaffected by the virtualized applications **124**. Using virtualized applications **124** provides the flexibility to add or remove these applications dynamically either as a required app pushed down by the management service **120** and retrieved by the management agent **212**, or retrieved on-demand and downloaded to the managed VM **122** upon a user request through a user interface or console provided by the management service **120**.

(59) The mounting of application virtual disk volumes **224a . . . 224n** received as part of one or more virtual application packages **251** can occur within the managed VM **122** itself. This prevents the need for any operation on the host device **106** to add or remove virtual storage devices. This implies no dependency on the desktop hypervisor APIs allowing this to work for any host desktop hypervisor **203** product including Fusion, Workstation, Player, Hyper-V Parallels, KVM, and VirtualBox, among others. For virtual application packages **251** downloaded previously or cached, no further download of data is required, and the mount operation will occur near instantaneously and can occur on demand. Layered or virtualized applications **124** can also co-exist alongside other non-layered or non-virtualized VM installed applications, all delivered and managed by the management service **120** seamlessly. An administrator can create new virtual application packages **251** for virtualized applications **124** when desired through the console user interface of the management service **120**.

(60) Each application virtual disk volume **224** can include a virtual hard disk such as a .vhd file or another type of virtual hard disk file. The managed virtual machine **122** can include instructions that enable the managed virtual machine **122** to mount the .vhd file of the application virtual disk volume **224** on demand, such as when a launch request is detected for a virtualized application **124**. For example, the application volume driver can enable the .vhd file of the application virtual disk volume **224** to be mounted to launch the virtualized application **124**. The managed virtual machine **122** can unmount the .vhd file of the application virtual disk volume **224** once the virtualized application **124** is closed, or the application virtual disk volume **224** can remain mounted.

(61) A user of the host device **106** can navigate through a user interface provided by the host operating system **151**, and can select an icon for an application corresponding to the virtualized application **124**. This can transmit an application launch request to the host operating system **151**. A launch agent such as the virtual application agent **215** can detect and intercept the application launch request. The virtual application agent **215** can utilize communications received through the IPC channel **211** to detect and intercept the application launch request. The host managed VM service **209** can intercept the application launch request and forward the request or an indication of the application launch request through the IPC channel **211**.

(62) The virtual application agent **215** can then mount the application virtual disk volume **224a** corresponding to the requested application. The virtual application agent **215** can map a filename or another parameter within the application launch request to a particular application virtual disk volume **224a**. Once the application virtual disk volume **224a** is mounted, the virtualized application **124a** can be generated for display on the host device **106**. From the user's perspective it can appear that the virtualized application **124a** has launched within the host operating system **151**, while it is actually executed using the guest operating system **221** kernel. In other examples, the user can access a user interface of the guest operating system **221** and select the appropriate application icon. In that case, the virtual application agent **215** can identify or intercept the application launch request within the guest operating system **221** and mount the application virtual disk volume **224a** corresponding to the requested application to launch the virtualized application **124a**. The application virtual disk volume **224a** can include a static virtual disk volume file that contains one or more applications that can be mounted in a managed virtual machine **122** as a read-only virtual disk. As a result, the application virtual disk volume **224a** is not written to or modified by mounting the application virtual disk volume **224a** and launching the virtualized application **124a** or virtualized applications **124** therefrom.

(63) One or more writeable volume **134** can be used to redirect all write operations to a separate volume from the guest OS volume **236** and the application virtual disk volumes **224a . . . 224n**. The writeable volume **134** can be specific to one or multiple virtualized applications **124**, and can be volatile or nonvolatile based on the use case. Using virtualized application writable volumes **134** can protect the underlying guest OS **221** file system from change. Writeable volumes **134** can be utilized on managed VMs **122** as a file container or information silo for one or more virtualized applications **124a . . . 124n**. A writeable volume **134** can isolate data from other applications within the managed virtual machine **122**, whether virtualized or installed. As can be understood, the managed virtual machine **122** further isolates all application therein from personal applications **251a . . . 251n** of the host environment. Write actions from virtualized applications **124a . . . 124n** can be handled in a number of ways. For example, the guest OS **221** kernel and the application volumes driver **222** can direct write operations according to the following use cases: 1. No virtual application specific writeable volumes **134**: All writes from virtualized applications **124a . . . 124n** can be written directly to the guest OS file system. 2. Single nonvolatile application writeable volume **134**: All writes are redirected to a single writeable volume **134** for multiple virtualized applications **124a . . . 124n**, which can include all or a specified set of siloed applications from the virtualized applications **124a . . . 124n**. The set of siloed virtualized applications **124** can be mounted and launched from a single application virtual disk volume **224**, or from multiple different application virtual disk volumes **224**. 3. Single volatile application writeable volume **134**: All writes are redirected to a single writeable volume **134** for multiple virtualized applications **124a . . . 124n**, which can include all or a specified set of siloed applications from the virtualized applications **124a . . . 124n**, but do not persist between managed VM **122** or host device **106** reboots or application restarts (analogous to a ram disk) The set of siloed virtualized applications **124** can be mounted and launched from a single application virtual disk volume **224**, or from multiple different application virtual disk volumes **224**. 4. Multiple individual nonvolatile writeable application writeable volumes **134**: Each of the virtualized applications **124a . . . 124n** writes to its

own specific application writeable volume **134**. Writes are not visible or accessible to other applications. 5. Multiple individual volatile writeable application writeable volumes **134**: Each of the virtualized applications **124a . . . 124n** writes to its own specific application writeable volume **134**. Writes are not visible or accessible to other applications, and do not persist between managed VM **122** or host device **106** reboots or application restarts.

(64) Use case 1 is low in complexity, but provides no protection to the guest OS **221** file system. This is analogous to a monolithic persistent virtual desktop infrastructure (VDI) scenario. Use cases 2 and 3 protect the underlying guest OS **221** file system from change but allows virtualized applications **124a . . . 124n** to share a common view of the aggregated file system and registry. This isolates the guest OS **221** file system from the virtualized applications **124a . . . 124n** but allows all or a specified subset of virtualized applications **124a . . . 124n** to share files and registry keys between each other. Use cases 4 and 5 are the fully isolated modes where writes are private to each virtualized application **124**. The volatile use cases can be used to ensure the virtualized application **124** starts in a consistent predetermined state every time, or maybe used to prevent unbounded usage of disk space. This can be utilized for testing, and scenarios where persistent files and customized configuration information is not required.

(65) A mixture of these modes can be used depending on the needs of each virtualized application **124**. For instance, a certain virtualized application **124** that writes confidential data can utilize an individual application-specific writeable volume **134**, while another set of virtualized applications **124** uses a shared writeable volume **134**. A volatile writeable volume can provide a workaround for a legacy virtualized application **124** that tends degrade in performance over time as its cache builds up, while other virtualized applications **124** use nonvolatile writeable volumes **134**. While traditional virtualized application **124** deployments can access the application remotely, this solution can deliver the virtualized application **124** for installation on the managed virtual machine **122**, which uses the host hardware **206** of the host device **106**. As a result, the user can remain productive while offline or outside of an enterprise intranet.

(66) The virtualized applications **124** can include implementations using VMware® AppVolumes, Citrix® App Layering, Liquidware® Flexapp, or container programs such as Docker®.

(67) The managed VM **122** can include a number of security features that can protect and isolate the guest environment of the managed VM **122** from the host environment. This can include automatic provisioning of a vTPM device using partial or full VM encryption and a credential generation model which can be used by the guest to protect Bitlocker® and other keys and credentials using a vTPM device. Management-service-enforced configurations such as policies **113**, profiles **114**, and restrictions, can be enforced based on the managed VM **122**. The managed VM **122** can be signed to ensure authenticity and integrity. The managed VM **122** can include a signature and a signed parameter file. The signed parameter file can include the managed VM configuration file, or all or a subset of the parameters from the managed VM configuration file. A managed VM user interface locking feature can prevent a user of the host device **106** from changing settings of the managed VM **122** from a host desktop hypervisor **203** user interface. Managed VM configuration file parameters can be signed to prevent file tampering with respect to all, or a predetermined set, of parameters of the managed VM configuration file. This can include an option to automatically repair to a default set of parameters if the guest management agent **212**, the host managed VM service **209**, or the host desktop hypervisor **203** detect that the managed VM configuration file parameters have been altered.

(68) The host managed VM service **209** can communicate with the guest managed VM service **210** through an inter-process communication (IPC) channel **211**. The IPC channel **211** can provide bidirectional guest-host and host-guest communications. In some examples, the host desktop hypervisor **203** can include a method of exchanging data that can be utilized to provide the IPC channel **211**. The host managed VM service **209** can include an application that runs in the user context ensuring no code runs in system context of the host device **106** and the host OS **151**. The

host managed VM service **209** can utilize host desktop hypervisor **203** APIs, such as REST APIs, VMREST APIs, and others. Host desktop hypervisors **203** can include, VMware® Workstation, VMware Fusion® and other desktop hypervisors.

(69) The IPC channel **211** can be achieved in a number of ways. For example, one type of IPC channel **211** can include a host-only network that involves provisioning a network device into the managed VM **122** and building the IPC channel **211** using that provisioned network device. This type of IPC channel **211** can support and enable communications using standard TCP/IP, UDP/IP, and other web-based protocols such as REST or gRPC.

(70) Another type of IPC channel **211** can include a custom socket-based service using a remote procedure call (RPC) backchannel of the host desktop hypervisor **203**. This RPC backchannel service can use vSockets, other VM communication interface (VMCI) socket capabilities.

(71) Yet another type of IPC channel **211** can include an IPC backchannel of the host desktop hypervisor **203** that operates using vmrun and rpctool. This third type of IPC channel can allow any process running in the user context of the host device **106** to send and receive variables. This type of IPC channel **211** can be limited to passing key value pairs in an asynchronous variable passing manner.

(72) The host desktop hypervisor **203** can be modified to support managed VMs **122**. Partial encryption support, including VM protected settings encryption support can be included. For example, the host desktop hypervisor **203** can add a VTPM device to the managed VM **122** using partial encryption and automatic key generation if an auto-add vTPM option is set to software in the managed VM configuration file. The host desktop hypervisor **203** can also support verification that a restricted or protected subset of the parameters of the VM configuration file are unaltered prior to running the managed VM **122**, while allowing the unrestricted subset of the parameters of the VM configuration file to be changed. The protected subset can include all or some of the managed VM configuration parameters.

(73) The managed VM **122** can enforce rules based on profiles **114** or policies **113**, and in response to commands from the management service **120**. For example, the guest managed VM service **210** can identify a failure to comply with profiles **114**, policies **113**, and other statuses that affect the compliance status **115**. Even if the host device **106** is unable to access the management service **120**, the guest managed VM service **210** can perform predetermined remedial and other actions based on a failure to comply with a particular policy **113** or profile **114**. The guest managed VM service **210** can also perform actions in response to commands received from the management service **120**.

(74) The guest managed VM service **210** can transmit a command to perform a certain data modification action through the IPC channel **211** to the host managed VM service **209**. The action can include enterprise wipe, which can cause all data, including applications, and all enterprise resources **119** to be removed from the managed VM **122**. In addition, the managed VM **122** can be removed from the host device **106** entirely. This can include a command to delete specified data from the managed VM **122**, or to delete the managed VM **122** from a file system of the host device **106**. The host managed VM service **209** can suspend or stop execution of the managed VM **122** using the host desktop hypervisor **203**, and then delete the managed VM **122** or specified data. If only a subset of data from the managed VM **122** is to be removed, the managed VM **122** can be restarted thereafter.

(75) The guest managed VM service **210** can also transmit a lock or unlock command for the managed VM **122** through the IPC channel **211** to the host managed VM service **209**. The host managed VM service **209** can use the host desktop hypervisor **203** to prevent the managed VM **122** from executing. In some cases, the lock command can also prevent a user of the host device **106** from logging into the host desktop hypervisor **203**, or prevent alteration of certain settings through a user interface of the host desktop hypervisor **203**. The guest managed VM service **210** can also cause the host managed VM service **209** to implement networking restrictions for network data originating from the managed VM **122**.

(76) FIG. 3 shows a flowchart performed by components of the networked environment **100**.

Generally, the flowchart illustrates how components of the networked environment **100** work in concert to package and provision a managed VM **122** on a host device **106**.

(77) At step **303**, an administrator can distribute or deploy a managed VM **122** to a host device **106** using a management service **120**, a PCLM, SCCM, AD, UEM, MDM, or other management services, manually, or using a script. The managed VM **122**, or a distributable package to deploy the managed VM **122**, can also be provided through a removable drive connected to the host device **106**, a file share accessed by the host device **106**, a FTP server, or similar systems. The provisioning tool can provision a managed VM **122** on the host device **106** using the managed VM **122** package. The managed VM **122** package, the host managed VM service **209**, or the host desktop hypervisor **203** can include a managed VM provisioning tool. The managed VM provisioning tool can initiate and implement a provisioning process that installs the managed VM **122**. The provisioning process can use a managed VM **122** package to install, configure, and launch a managed VM **122**.

(78) In some cases, the initial managed VM **122** package can include one or more virtualized applications **124**, including virtual disk files to create corresponding application virtual disk volumes **224** and instructions for a writeable volume **234** to create or use for each virtualized application **124**. However, in other examples, the managed VM **122** itself is installed and configured with initial settings, and the management agent **212** can then retrieve a user-specific set of configurations associated with a user or user group.

(79) At step **306**, the managed VM **122** can receive a command to provision the managed VM **122** with a virtualized application **124**. For example, the management agent **212** can check in with the management service **120** and retrieve a command from a command queue for the managed VM **122**. The command can include a command to download data of the virtualized application **124** from a network location. The network location can include a uniform resource link (URL), uniform resource identifier (URI), or universal naming convention (UNC) file path among other identifiers. The virtualized application **124** data can include a file such as a virtual disk file. The virtual disk file can include one or more virtualized applications **124**. The virtualized application **124** data can also include configuration instructions for how to provision and configure the virtualized application **124** on the managed VM **122**.

(80) At step **309**, the managed VM **122** can store an application virtual disk volume **224** for the virtualized application **124**. For example, the management agent **212** can use the virtual disk file to create the application virtual disk volume **224**. The management agent **212** can create the application virtual disk volume **224** as a read-only volume within the host hardware **206** that is allocated to the managed VM **122**. In some examples, the application virtual disk volume **224** can include a set of virtualized applications **124**. In other examples, the application virtual disk volume **224** can be specific to a single virtualized application **124**.

(81) At step **312**, the managed VM **122** can determine whether to create a writeable volume **234**. Since the application virtual disk volume **224** can be read-only, write operations cannot be directed to that volume. Rather, write operations can be directed to the guest OS volume **236** or a specific writeable volume **234** that is used to silo write operations for one or more virtualized applications **124**. For example, the management agent **212** can check the policies **113**, profiles **114**, and other configuration instructions to identify where write operations are to be performed for the virtualized application **124**. The write operation instructions can be user specific, device specific, managed VM specific, or application specific for the virtualized application **124** or a specified group of virtualized applications **124**.

(82) The configuration instructions can indicate that a writeable volume **234** is to be created for the virtualized application **124**. In some cases, all virtualized applications **124** in the managed VM **122** can share a single writeable volume **234**. In other examples, each virtualized application **124** utilizes a corresponding application-specific writeable volume **234**. Alternatively, the configuration

instructions can also include whitelisted applications with which the virtualized application **124** is permitted to share a writeable volume **234**, blacklisted applications with which the virtualized application **124** is prohibited from sharing a writeable volume **234**, and so on. The management agent **212** can check a current listing and configuration of writeable volumes **234** and determine whether a writeable volume **234** corresponds to the writeable volumes requirements indicated in the configuration instructions. If a writeable volume **234** is to be created based on configuration instructions and the currently existing writeable volumes **234**, then the process moves to step **315**. Otherwise, the process can move to step **318**.

(83) At step **315**, the management agent **212** or other instructions executed on the managed VM **122** can create a writeable volume **234** according to the configuration instructions. The writeable volume **234** can be volatile or nonvolatile according to the configuration instructions. The writeable volume **234** can be used by a single virtualized application **124** or multiple virtualized applications **124** according to the configuration instructions. Once the writeable volume **234** is created, disk and registry write operations by the virtualized application **124** that would otherwise be made to the local guest OS volume **236** can be redirected to the writeable volume **234**. This can prevent changes to the guest OS volume **236** or other installed applications such as VM installed applications **218** and personal applications **251** installed to the host device **106**.

(84) At step **318**, the management agent **212** or other instructions executed on the managed VM **122** can configure the virtual application agent **215** or another launch redirection agent. The management agent **212** or the virtual application agent **215** can request or otherwise facilitate the guest managed VM service **210** to instruct the host managed VM service **209** via the IPC channel **211** to intercept application launch requests and provide an indication of the application launch request through the IPC channel **211**. In some examples, the virtual application agent **215** or the management agent **212** can request or otherwise facilitate the guest managed VM service **210** to instruct the host managed VM service **209** via the IPC channel **211** to create an application launch icon that links to instructions for the host managed VM service **209**. This can provide an indication of the application launch request through the IPC channel **211**.

(85) FIG. **4** shows a flowchart performed by components of the networked environment **100**. Generally, the flowchart illustrates how components of the networked environment **100** launch and execute a virtualized application **124** within a managed VM **122** on a host device **106**.

(86) At step **403**, the virtual application agent **215** can utilize communications received via the guest managed VM service **210** through the IPC channel **211** to detect the application launch request. The host managed VM service **209** can intercept the application launch request and forward the request or an indication of the application launch request through the IPC channel **211**.

(87) At step **406**, the virtual application agent **215** can mount the application virtual disk volume **224**. The virtual application agent **215** can identify the virtualized application **124** associated with or specified by the application launch request. The virtual application agent **215** can utilize the application volumes driver **222** to mount the application virtual disk volume **224** corresponding to the requested virtualized application **124**.

(88) In step **409**, the virtual application agent **215** can launch the requested virtualized application **124** within the managed VM **122**. The virtualized application **124** can be executed using the guest OS **221** kernel of the managed VM **122**. While the virtualized application **124** can be executed within the managed VM **122**, it can appear from the user perspective the same as any application launched from the host OS **151** and executed using the host OS **151** kernel. The virtualized application **124** can also include applications that do not create a user interface.

(89) In step **412**, the virtualized application **124** can store write operations within the managed VM **122**. The virtualized application **124** can be configured to direct write operations to the writeable volume **234**. The virtual application agent **215**, the application volumes driver **222**, the guest OS **221** kernel, or other instructions within the managed VM **122** can be used to direct or redirect write operations for the virtualized application **124** to the writeable volume **234**. This can protect the

guest OS **221**, the VM installed applications **218**, and the guest OS volume **236** and file system from being affected by the virtualized application **124**. If no writeable volume **234** is to be utilized, the virtualized application **124** can write operations to the guest OS volume **236**, much like the VM installed applications **218**. This can still protect the host OS **151** from being affected by the virtualized application **124**. The virtualized application **124** can also access and read from the data written to the writeable volume **234**, including data written by other virtualized applications **124** that are directed and permitted to write to the writeable volume **234**.

(90) The flowchart(s) and sequence diagram(s) show examples of the functions and operation of the components described herein. The components described herein can be embodied in hardware, software, or a combination of hardware and software. If embodied in software, each element can represent a module or group of code that includes program instructions to implement the specified logical function(s). The program instructions can be embodied in the form of, for example, source code that includes human-readable statements written in a programming language or machine code that includes machine instructions recognizable by a suitable execution system, such as a processor in a computer system or other system. If embodied in hardware, each element can represent a circuit or several interconnected circuits that implement the specified logical function(s).

(91) The computing environment **103** can include at least one processing circuit. Such a processing circuit can include, for example, one or more processors and one or more storage or memory devices that are coupled to a local interface. The local interface can include, for example, a data bus with an accompanying address/control bus or any other suitable bus structure. Similarly, the host devices **106** can each include at least one processing circuit. Such a processing circuit can include, for example, one or more processors and one or more storage or memory devices that are coupled to a local interface.

(92) The storage or memory devices can store data or components that are executable by the processors of the processing circuit. For example, the management service **120** and/or other components can be stored in one or more storage devices and be executable by one or more processors in the networked environment **100**. Similarly, the agents, services, applications and/or other components described herein can be stored in one or more storage devices and be executable by one or more processors in the host device **106**.

(93) The management service **120** and/or other components described herein can be embodied in the form of hardware, software components that are executable by hardware, or as a combination of software and hardware. If embodied as hardware, the components described herein can be implemented as a circuit or state machine that employs any suitable hardware technology. The hardware technology can include, for example, one or more microprocessors, discrete logic circuits having logic gates for implementing various logic functions upon an application of one or more data signals, application specific integrated circuits (ASICs) having appropriate logic gates, programmable logic devices (e.g., field-programmable gate arrays (FPGAs), and complex programmable logic devices (CPLDs)).

(94) Also, one or more of the components described herein that include software or program instructions can be embodied in any non-transitory computer-readable medium for use by or in connection with an instruction execution system such as, a processor in a computer system or other system. The computer-readable medium can contain, store, and/or maintain the software or program instructions for use by or in connection with the instruction execution system.

(95) A computer-readable medium can include a physical media, such as, magnetic, optical, semiconductor, and/or other suitable media. Examples of a suitable computer-readable media include, but are not limited to, solid-state drives, magnetic drives, or flash memory. Further, any logic or component described herein can be implemented and structured in a variety of ways. For example, one or more components described can be implemented as modules or components of a single application. Further, one or more components described herein can be executed in one computing device or by using multiple computing devices.

(96) Further, any logic or applications described herein, including the management service **120** and/or other components can be implemented and structured in a variety of ways. For example, one or more applications described can be implemented as modules or components of a single application. Further, one or more applications described herein can be executed in shared or separate computing devices or a combination thereof. For example, a plurality of the applications described herein can execute in the same computing device, or in multiple computing devices. Additionally, terms such as “application,” “service,” “system,” “engine,” “module,” and so on can be used interchangeably and are not intended to be limiting.

(97) The above-described examples of the present disclosure are merely possible examples of implementations set forth for a clear understanding of the principles of the disclosure. Many variations and modifications can be made without departing substantially from the spirit and principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

Claims

1. A method implemented by instructions executed by at least one computing device, the method comprising: receiving, at a host device from a management service, a managed virtual machine enrolled with the management service and a set of compliance rules to be enforced on the managed virtual machine; executing, within the host device comprising a host operating system, the managed virtual machine, wherein the managed virtual machine comprises a guest operating system and a management agent that communicates with the management service; storing, within the managed virtual machine, a virtualized application drive comprising a virtualized application; receiving an application access request that is generated by input on a user interface displayed by the host operating system of the host device; intercepting the application access request by a host managed virtual machine (VM) service executing in the host operating system and forwarding the application access request from the host managed VM service to a guest managed VM service executing in the guest operating system of the managed virtual machine, wherein the guest managed VM service provides the application access request to a launch agent executing within the guest operating system of the managed virtual machine; launching, by the launch agent, the virtualized application in the guest operating system by accessing the virtualized application drive stored in the managed virtual machine; and enforcing the set of compliance rules received from the management service on the managed virtual machine and the virtualized application by the management agent.
2. The method of claim 1, wherein the host device is not enrolled with the management service, and a host OS file system lacks a component that communicates with the management service.
3. The method of claim 1, wherein the application access request is forwarded through an inter-process communication (IPC) channel established between the host managed VM service and the guest managed VM service.
4. The method of claim 1, further comprising: storing, within the managed virtual machine, a write operation of the virtualized application.
5. The method of claim 4, further comprising: creating, within the managed virtual machine, a writeable volume for write operations of the virtualized application, wherein the write operation is stored within the writeable volume.
6. The method of claim 5, wherein the writeable volume is a multiple-application writeable volume utilized by a plurality of virtualized applications corresponding to a plurality of virtual application drives stored within the managed virtual machine and within hardware of the host device.
7. The method of claim 5, wherein the writeable volume is a single-application writeable volume utilized by a single one of a plurality of virtualized applications corresponding to a plurality of virtual application drives.

8. The method of claim 1, wherein in response to a command received by the management agent from the management service, the guest managed VM service transmits to the host management VM service a command to perform an enterprise wipe action that causes the managed virtual machine including all data, applications and enterprise resources to be deleted.
9. A non-transitory computer-readable medium embodying instructions executable by at least one computing device wherein the instructions, when executed, cause the at least one computing device to at least: receive, at a host device from a management service, a managed virtual machine enrolled with the management service and a set of compliance rules to be enforced on the managed virtual machine; execute, within the host device comprising a host operating system, the managed virtual machine, wherein the managed virtual machine comprises a guest operating system and a management agent that communicates with the management service; store, within the managed virtual machine, a virtualized application drive comprising a virtualized application; receive an application access request that is generated by input on a user interface displayed by the host operating system of the host device; intercept the application access request by a host managed virtual machine (VM) service executing in the host operating system and forwarding the application access request from the host managed VM service to a guest managed VM service executing in the guest operating system of the managed virtual machine, wherein the guest managed VM service provides the application access request to a launch agent executing within the guest operating system of the managed virtual machine; launch, by the launch agent, the virtualized application in the guest operating system by accessing the virtualized application drive stored in the managed virtual machine; and enforce the set of compliance rules received from the management service on the managed virtual machine and the virtualized application by the management agent.
10. The non-transitory computer-readable medium of claim 9, wherein the host device is not enrolled with the management service, and a host OS file system lacks a component that communicates with the management service.
11. The non-transitory computer-readable medium of claim 9, wherein the application access request is forwarded through an inter-process communication (IPC) channel established between the host managed VM service and the guest managed VM service.
12. The non-transitory computer-readable medium of claim 9, wherein, when executed, the instructions further cause the at least one computing device to at least: store, within the managed virtual machine, a write operation of the virtualized application.
13. The non-transitory computer-readable medium of claim 12, wherein, when executed, the instructions further cause the at least one computing device to at least: create, within the managed virtual machine, a writeable volume for write operations of the virtualized application, wherein the write operation is stored within the writeable volume.
14. The non-transitory computer-readable medium of claim 13, wherein the writeable volume is a nonvolatile writeable volume utilized by at least one of a plurality of virtualized applications corresponding to a plurality of virtual application drives stored within the managed virtual machine and within hardware of the host device.
15. The non-transitory computer-readable medium of claim 13, wherein the writeable volume is a volatile writeable volume utilized by at least one of a plurality of virtualized applications corresponding to a plurality of virtual application drives stored within the managed virtual machine and within hardware of the host device.
16. A system, comprising: at least one processor; and instructions accessible by the at least one processor, wherein when executed the instructions cause the system to at least: receive, at a host device from a management service, a managed virtual machine enrolled with the management service and a set of compliance rules to be enforced on the managed virtual machine; execute, within the host device comprising a host operating system, the managed virtual machine, wherein the managed virtual machine comprises a guest operating system and a management agent that communicates with the management service; store, within the managed virtual machine, a

virtualized application drive comprising a virtualized application; receive an application access request that is generated by input on a user interface displayed by the host operating system of the host device; intercept the application access request by a host managed virtual machine (VM) service executing in the host operating system and forwarding the application access request from the host managed VM service to a guest managed VM service executing in the guest operating system of the managed virtual machine, wherein the guest managed VM service provides the application access request to a launch agent executing within the guest operating system of the managed virtual machine; launch, by the launch agent, the virtualized application in the guest operating system by accessing the virtualized application drive stored in the managed virtual machine; and enforce the set of compliance rules received from the management service on the managed virtual machine and the virtualized application by the management agent.

17. The system of claim 16, wherein the host device is not enrolled with the management service, and a host OS file system lacks a component that communicates with the management service.

18. The system of claim 16, wherein the application access request is forwarded through an inter-process communication (IPC) channel established between the host managed VM service and the guest managed VM service.

19. The system of claim 16, wherein, when executed, the instructions further cause the system to at least: store, within the managed virtual machine, a write operation of the virtualized application.

20. The system of claim 19, wherein, when executed, the instructions further cause the system to at least: create, within the managed virtual machine, a writeable volume for write operations of the virtualized application, wherein the write operation is stored within the writeable volume.
